Reg. No. | | | | | | | | | | |

[12EE831]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

ELECTRICAL POWER QUALITY

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Distinguish between the types of transients that occur in power systems with sketches, cause of their occurance and how they can be mitigated. **07**

   b. Explain the power quality terms: CBEMA curve, Distributed Generation, Linear Load, non-linear load, equipment grounding conductor. **07**

2. a. Describe with a neat diagram the voltage sag solution alternatives. **06**

   b. List out all the causes of a voltage sag disturbance in Power systems with relevant sketches. **08**

3. a. Explain the principles of over voltage protection. **07**

   b. Explain the different modes of capacitor switching, their significance and when they are put into use. **07**

4. a. What are the effects of harmonics? Bring out the differences between voltage and current harmonic with an example. **06**

   b. What is the difference between true power factor and displacement power factor? In the case of a particular industrial customer the following readings were observed. **08**

RMS in pu	Frequency 50Hz	Frequency of 150Hz	Frequency of 750Hz
V	$1\angle 0$	$0.33\angle 30°$	$0.7\angle 10°$
I	$1\angle -60°$	$1\angle 80°$	$0.2\angle -30°$

Find the true power factor and displacement power factor. In the above problem which power factor would the customer prefer? Why?

5. a. Explain the overall procedure for utility system harmonic evaluation **07**

   b. Write a note on: **07**

   i. Harmonic filters

   ii. Standards of harmonics

6.	**a.**	Explain the general approach to power quality state estimation.	**07**
	b.	Write in detail about power quality in distribution planning.	**07**
7.	**a.**	List out the main power quality issues affected by distributed generation and give a brief note on each.	**07**
	b.	Describe briefly the special applications problems that do not arise in the usual radial services arrangement.	**07**
8.	**a.**	What is power quality monitoring? Why is it required?	**07**
	b.	Distinguish the two types of power quality data analysis.	**07**